Assumptions Log

This document records all assumptions made during the EMS Readiness Optimization project.

Data Assumptions

A1: Crash Data Representativeness

Assumption: Historical crash data (2013-2026) is representative of future crash patterns.
Rationale: Long time series provides stable estimates of spatial and temporal patterns.
Risk: Major changes in traffic patterns (e.g., congestion pricing) could alter future demand.
Mitigation: Include sensitivity analysis on demand levels.

A2: Location Accuracy

Assumption: Crash latitude/longitude coordinates are sufficiently accurate for demand estimation.
Rationale: NYPD geocoding process should provide reasonable accuracy for most incidents.
Risk: Some records may have missing or inaccurate coordinates.
Mitigation: Filter records with missing coordinates; aggregate to grid cells.

A3: Firehouse Availability

Assumption: All Manhattan firehouses are available as potential EMS staging locations.
Rationale: Study explores optimal staging locations; operational constraints addressed separately.
Risk: Some firehouses may not have space or infrastructure for EMS staging.
Mitigation: Document as limitation; can add constraints in future iterations.

Model Assumptions

A4: Travel Time Model

Assumption: Travel times can be approximated using Euclidean distance with a scaling factor or simplified network distances.
Rationale: Full routing adds complexity without fundamentally changing optimization insights.
Risk: May over/underestimate actual response times in some areas.
Mitigation: Sensitivity analysis on travel time multipliers.

A5: Single Unit Response

Assumption: Each crash incident requires exactly one EMS unit.
Rationale: Simplifies simulation while capturing core dispatch dynamics.
Risk: Major incidents may require multiple units.
Mitigation: Model captures most incidents; note as limitation.

A6: Homogeneous Service Time

Assumption: Service time (on-scene + transport) follows a single distribution.
Rationale: Data on actual service times not available; use literature estimates.
Risk: Service times may vary by incident severity.
Mitigation: Sensitivity analysis on service time parameters.

A7: Immediate Dispatch

Assumption: EMS units are dispatched immediately upon incident occurrence.
Rationale: Focuses on travel time rather than dispatch delay.
Risk: Real systems have dispatch processing time.
Mitigation: Can add dispatch delay parameter if needed.

Geographic Assumptions

A8: Manhattan Boundary

Assumption: Incidents and firehouses are classified by whether they fall within Manhattan borough boundary.
Rationale: Standard geographic classification for NYC.
Risk: Boundary edges may have some ambiguity.
Mitigation: Use official borough boundary polygons.

A9: CBD Definition

Assumption: CBD is defined as the MTA Congestion Relief Zone boundary.
Rationale: Provides consistent, official boundary for the high-density core.
Risk: Other CBD definitions exist; results depend on boundary choice.
Mitigation: Document specific boundary used; can test alternatives.

Operational Assumptions

A10: Firehouse Staging Capacity (Updated v1.3.0)

Assumption: ~~Any number of EMS units can stage at a single firehouse.~~ **Updated:** Maximum 2 EMS units per firehouse (default capacity constraint).
Rationale: Full capacity sensitivity analysis (cap 1–5) demonstrates that cap=2 is operationally realistic based on typical FDNY firehouse infrastructure, and matches or improves performance compared to cap=5 at K ≤ 40. At K=20, capacity never binds (all policies allocate ≤ 1 unit/firehouse). At K=40, cap=2 forces wider geographic dispersion (29 vs 24 firehouses for P2).
Risk: Some firehouses may accommodate more or fewer units depending on physical infrastructure.
Mitigation: Capacity sensitivity analysis (cap 1–5) quantifies performance impact. Results are robust across all tested capacity values.

A11: No Pre-emption

Assumption: Once an EMS unit is assigned to an incident, it completes service before taking new calls.
Rationale: Standard EMS operational policy.
Risk: Pre-emption for high-priority calls does occur.
Mitigation: Aligns with most common practice; note limitation.

A12: Manhattan Distance Approximation

Assumption: Manhattan (taxicab) distance provides an upper-bound approximation of actual road distance in a grid-based network; Haversine provides a lower bound.
Rationale: Manhattan's street grid is largely rectangular; true road distance falls between Haversine and Manhattan distance.
Risk: Diagonal avenues and non-grid areas (e.g., lower Manhattan) deviate from both approximations.

Mitigation: Compared both metrics side-by-side; P2 allocation is robust to metric choice (mean RT differs by <0.01 min in simulation).

A13: CBD-Focused Demand Weighting

Assumption: Applying a 3× demand weight to CBD precincts in the CBD-focused model is a reasonable multiplier to test CBD prioritisation.
Rationale: The 3× factor substantially up-weights CBD demand (from 56% to ~77% effective weight) without completely ignoring non-CBD.
Risk: A different multiplier (e.g., 2× or 5×) might yield different conclusions.
Mitigation: Results show that even with 3× weight, CBD RT does not improve (2.50 vs. 2.47 min), while non-CBD RT degrades dramatically (+159%). This confirms that the Manhattan-wide P2 is robust and any CBD-focused up-weighting is counterproductive.

A14: Spatially-Stratified Baseline (P0-spatial) (Added v1.3.0)

Assumption: The baseline uniform allocation policy (P0) distributes units across firehouses using latitude-based spatial stratification rather than arbitrary index-based ordering.
Rationale: Index-based allocation (original P0) assigned units to firehouses in database-order, producing geographically biased clusters. Latitude-based stratification divides Manhattan into equal-width latitude bands and round-robins units across bands, ensuring even north–south geographic coverage regardless of firehouse ordering in the dataset.
Risk: Latitude-only stratification does not account for east–west variation or local demand density.
Mitigation: Manhattan's elongated north–south geometry makes latitude the dominant spatial axis. P0-spatial is a baseline comparator; optimized policies (P1, P2) already account for demand. Simulation confirms P0-spatial reduces mean response time by ~61% vs. original P0 (3.17 vs. 8.08 min at K=20).

Assumption Review Schedule

Assumption	Review Date	Reviewer	Status
All	Initial	Project Lead	Documented

Last Updated: March 15, 2026